Final Project

STQM-342-001

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Wage Determinants and Gender Pay Gap Analysis (Part A)

Prepared for Internal Review by HR Research and Analytics

Executive Summary

This report examines whether gender-based wage differences exist within a large firm after accounting for relevant human-capital attributes. Using data from 50 employees and statistical modeling via IBM SPSS Modeler, hourly wage values were analyzed against four explanatory factors: years of higher education, work experience, age, and gender.

A multiple linear regression model was developed following an 85% training and 15% testing split. Stepwise selection identified only two statistically significant predictors of employee wage: Education and Male (gender indicator). These variables demonstrated nearly equal predictive strength, with Education showing importance of 0.51 and Male showing 0.49.

The final model estimates that male employees earn, on average, $4.50 more per hour than female employees with identical educational backgrounds. Model diagnostics indicate acceptable statistical validity for exploratory inference, though sample size is slightly below the recommended minimum. These findings suggest the presence of measurable gender-based pay differences within the sampled population.

Business Purpose

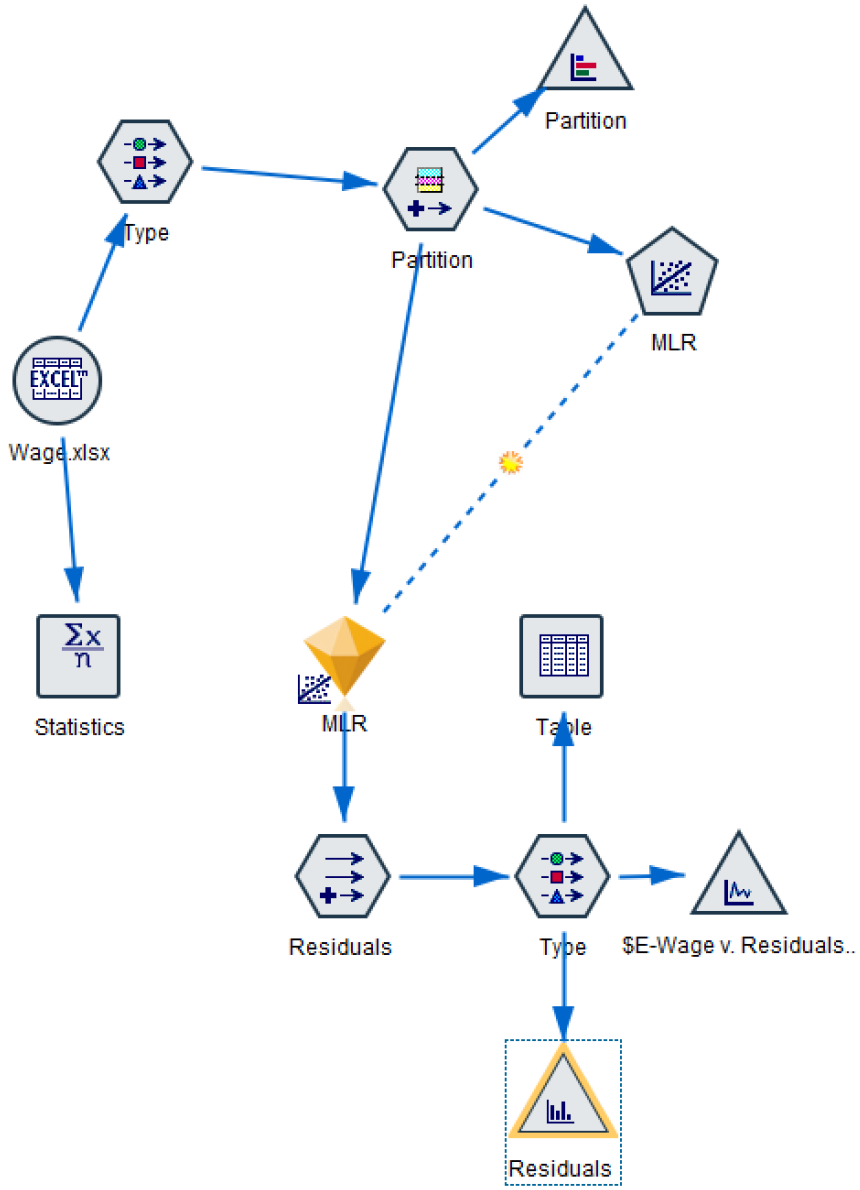
The firm requested an analytical review of wage outcomes to determine whether males receive higher compensation than females after accounting for education, experience, and age. The goal is to isolate statistically meaningful wage drivers and determine whether inequitable wage practices may be present.

Data Description and Processing

The original dataset contained 50 employee records with the following variables:

- Wage (hourly)
- Education (years of higher education)
- Experience (years of work experience)
- Age
- Male (1 = male, 0 = female)

The data were imported and defined in IBM-SPSS Modeler. Wage was assigned as the target field and all others as inputs. A partition split the data into 85% training (36 records) and 15% testing (14 records).



Correlation Analysis

A Pearson correlation matrix was generated to evaluate linear relationships among continuous variables and identify early multicollinearity concerns.

Male

Pearson Correlations

	Wage	Education	Experience	Age	Male
Wage	1.000/Perfect	0.420/Medium	0.291/Weak	0.102/Weak	0.379/Medium
Education	0.420/Medium	1.000/Perfect	-0.246/Weak	-0.114/Weak	0.368/Medium
Experience	0.291/Weak	-0.246/Weak	1.000/Perfect	0.421/Medium	0.095/Weak
Age	0.102/Weak	-0.114/Weak	0.421/Medium	1.000/Perfect	0.091/Weak
Male	0.379/Medium	0.368/Medium	0.095/Weak	0.091/Weak	1.000/Perfect

Key insights:

• Education and Wage: $r \approx 0.42$ (medium positive relationship)

• Male and Wage: $r \approx 0.38$ (medium positive relationship)

• Experience and Wage: $r \approx 0.29$ (weak)

• Age and Wage: $r \approx 0.10$ (very weak)

All predictors showed correlations below 0.70 with one another, indicating no strong multicollinearity concerns. These patterns suggested that education level and gender were most likely to have meaningful wage influence.

Sample Size Adequacy

For multiple regression, a widely accepted benchmark is:

Minimum required records ≈ 10 × number of inputs

With 4 inputs, the recommended minimum is:

$4 \times 10 = 40$ cases

The training partition contained 36 cases, slightly below the recommended threshold. Therefore, model findings remain exploratory and should be interpreted cautiously.

Regression Model Development

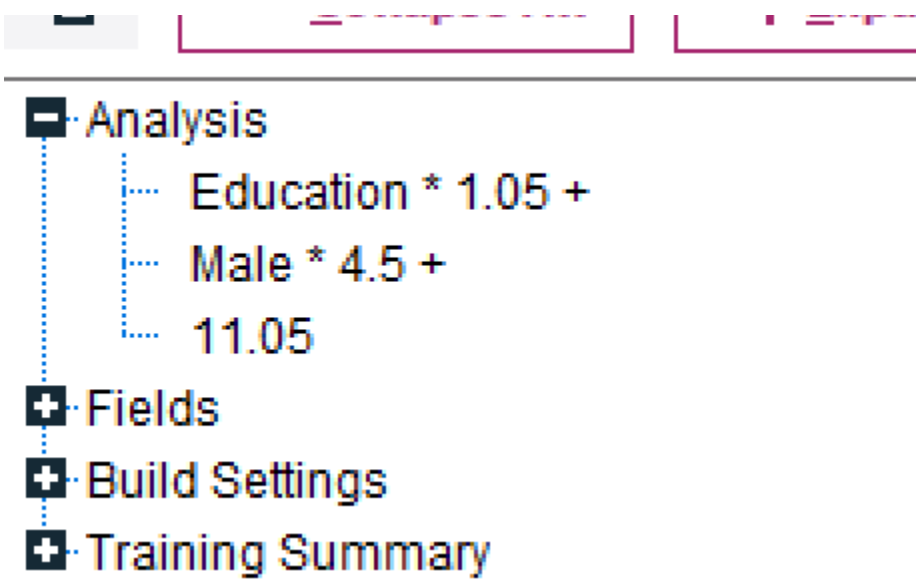
A stepwise multiple linear regression analysis was performed to determine the strongest wage predictors. Two models were generated:

Model 1:

Wage = 10.923 + 1.385(Education)

Model 2 (Final Model):

Wage = 11.047 + 1.050(Education) + 4.500(Male)



Coefficients

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	10.923	2.880		3.792	<.001		
	Education	1.385	.430	.483	3.220	.003	1.000	1.000
2	(Constant)	11.047	2.695		4.099	<.001		
	Education	1.050	.426	.366	2.466	.019	.894	1.119
	Male	4.500	1.862	.359	2.417	.021	.894	1.119

Excluded Variables

Model		Beta In	Collinearity Statistics VIF	Collinearity Statistics Minimum Tolerance
1	Experience	.255[b]	1.073	.932
	Age	.200[b]	1.056	.947
	Male	.359[b]	1.119	.894
2	Experience	.132[c]	1.298	.739
	Age	.150[c]	1.081	.830

b. Predictors in the Model: (Constant), Education

c. Predictors in the Model: (Constant), Education, Male

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Variance Proportions Education	Variance Proportions Male
1	1	1.946	1.000	.03	.03	
	2	.054	5.996	.97	.97	
2	1	2.555	1.000	.01	.01	.05
	2	.393	2.550	.05	.02	.90
	3	.052	7.003	.94	.96	.04

[Regression Model Output and Coefficients Table]

Interpretation of Coefficients

Intercept (11.047): Represents the predicted wage for a female employee (Male = 0) with zero education. Serves as a baseline reference rather than a literal scenario.

Education (1.050): Each additional year of higher education is associated with an estimated $1.05 increase in hourly wage, controlling for gender.

Male (4.500): Male employees are predicted to earn $4.50 more per hour than female employees with the same education level.

Statistical Significance

Education and Male both had p-values below 0.05, confirming that they are statistically significant predictors. Age and Experience did not significantly enhance the model once Education and Male were included and were excluded through the stepwise process.

Model Validity

The final model showed:

- $R = 0.591$
- $R^2 = 0.349$
- Adjusted $R^2 = 0.309$
- p-value < 0.001

[Model Summary and ANOVA Table]

Variables Entered/Removed

Model	Variables Entered	Variables Removed	Method
1	Education	.	Forward (Criterion: Probability-of-F-to-enter <= .050)
2	Male	.	Forward (Criterion: Probability-of-F-to-enter <= .050)

Double-click to activate

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.483[a]	.234	.211	5.608732	
2	.591[b]	.349	.309	5.247479	1.286

a. Predictors: (Constant), Education

b. Predictors: (Constant), Education, Male

ANOVA

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	326.154	1	326.154	10.368	.003[b]
	Residual	1069.568	34	31.458		
	Total	1395.721	35			
2	Regression	487.032	2	243.516	8.844	<.001[c]
	Residual	908.689	33	27.536		
	Total	1395.721	35			

b. Predictors: (Constant), Education

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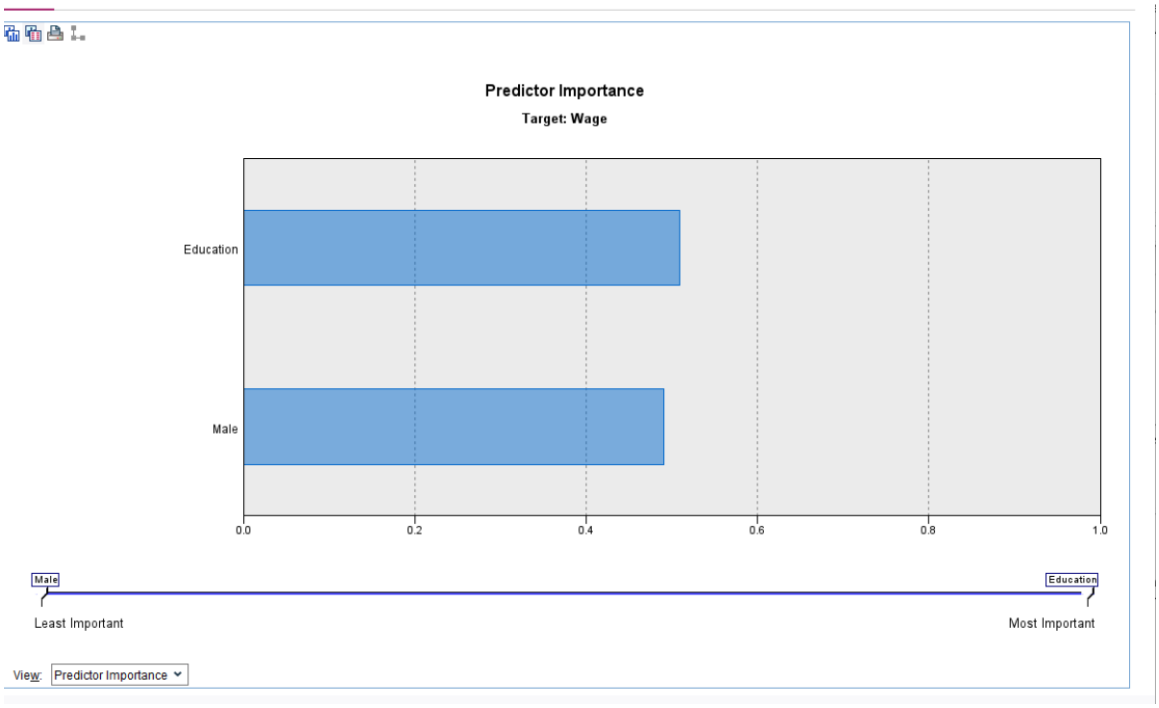
R^2 indicates that about 35% of the wage variance is explained by education and gender. Adjusted R^2 (31%) reflects good explanatory strength for a model with limited inputs and modest sample size.

Predictor Importance

IBM Modeler's predictor importance tool ranked Education as 0.51 and Male as 0.49, confirming nearly equal contribution to wage prediction strength.



Predicted Wage Scenarios

Profile:

Age = 40, Education = 10 years, Experience = 5 years

Male:

Wage = 11.047 + 1.050(10) + 4.500(1)

≈ $26.05/hr

Female:

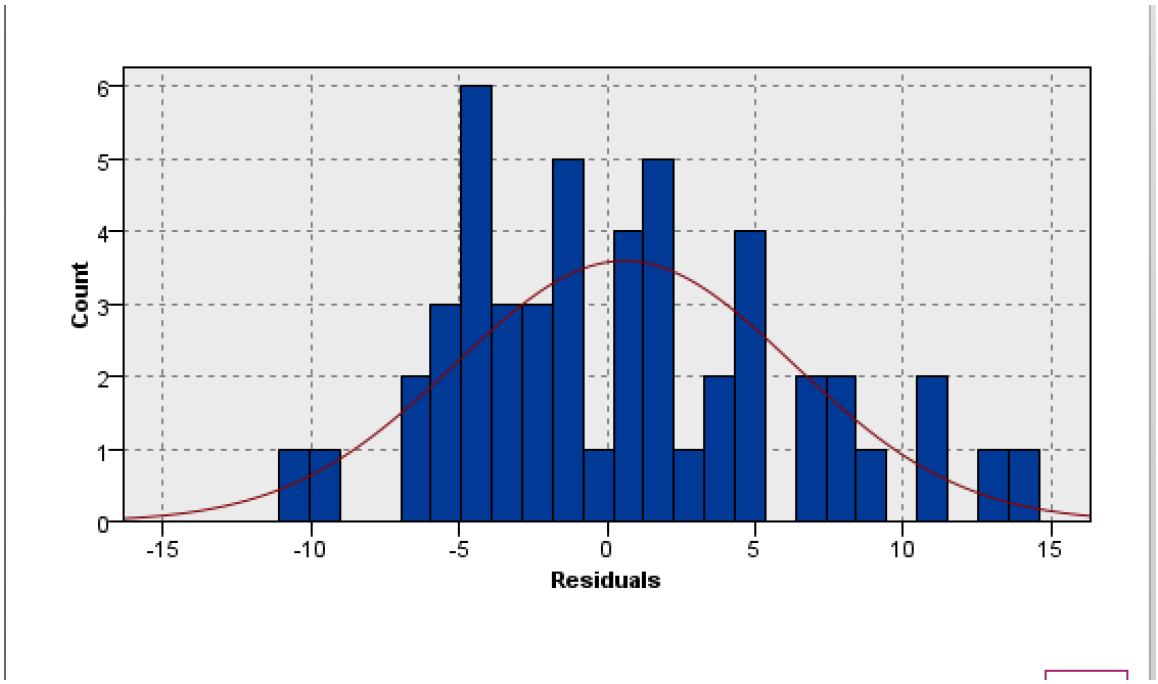
Wage = 11.047 + 1.050(10)

≈ $21.55/hr

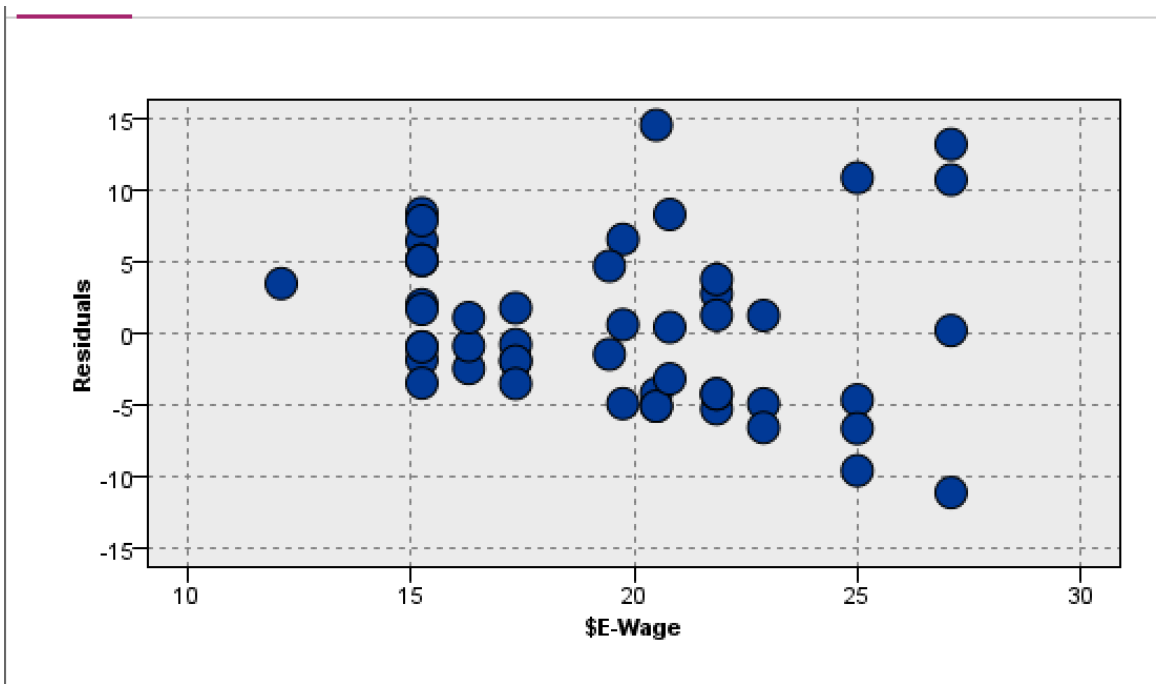
A $4.50/hr difference equates to approximately $9,360 per year assuming full-time hours.

Assumption and Diagnostic Checks

Normality: Residual histogram showed a roughly bell-shaped distribution centered near zero.



Linearity and Variance: Residual scatterplot displayed no funnel-shaped or curved patterns, indicating homoscedasticity and linearity are reasonable.



Outlier Review: Standardized residuals were within ±3, indicating absence of extreme outliers.

Residuals Statistics

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	15.24628	27.09498	19.69583	3.730308	36
Residual	-6.605313	13.235019	.000000	5.095346	36
Std. Predicted Value	-1.193	1.984	.000	1.000	36
Std. Residual	-1.259	2.522	.000	.971	36

```
End of job:  43 command lines  0 errors  0 warnings  2 CPU seconds
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Multicollinearity: VIF values averaged below 1.2, well within accepted cutoffs, and condition indices remained below 10.

Coefficients

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
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c. Predictors: (Constant), Education, Male

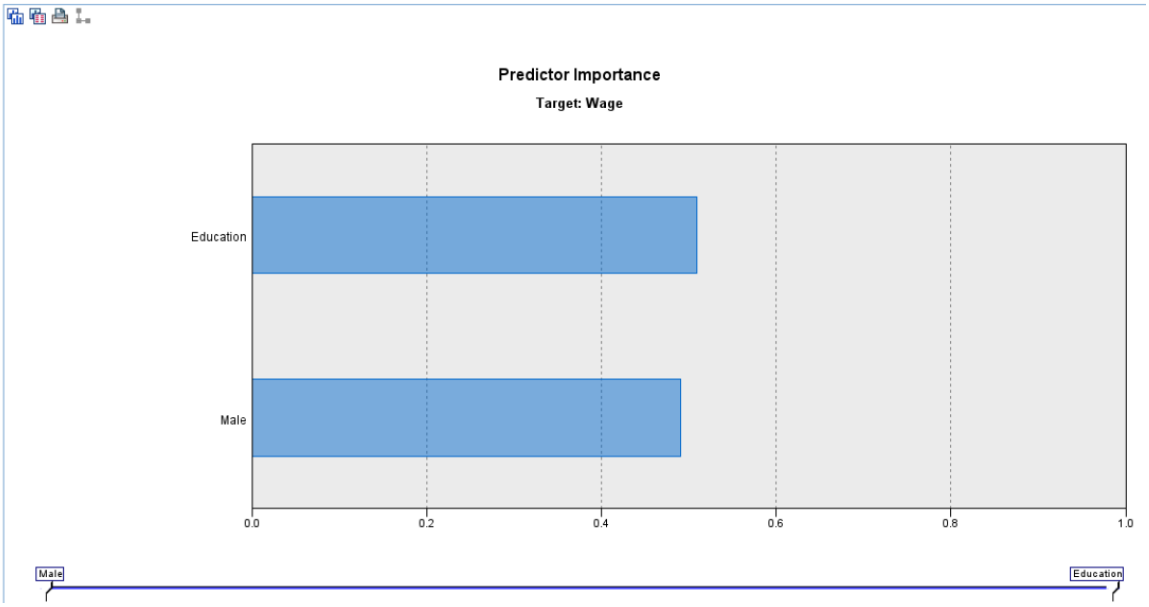
Model Comparison

Model 1 (Education Only):

Adjusted $R^2 = 0.211$

Model 2 (Education + Male):

Adjusted $R^2 = 0.309$



The inclusion of gender improved adjusted R^2 by nearly 10%, establishing the two-variable model as superior.

Sample Size Adequacy After Stepwise

After stepwise selection reduced predictors from four to two, the sample size recommendation becomes:

$2 \times 10 = 20$ cases

Training size of 36 comfortably exceeds this reduced minimum. The final model is reasonably supported by available data, though additional cases would further strengthen reliability.

Conclusion

This analysis demonstrates that education level and gender are significant determinants of hourly wage within the studied organization. The final regression model indicates that male employees earn, on average, $4.50 more per hour than female employees with identical educational backgrounds. Predictor importance scores confirm that gender is nearly as influential as education in determining wage outcomes.

Model validity checks confirm acceptable statistical conditions, and stepwise selection provided a more efficient model with stronger explanatory power. Although the dataset is slightly smaller than the recommended size for a four-input regression, final sample adequacy after stepwise reduction supports meaningful exploratory conclusions.

Given these results, leadership should consider additional review of compensation structures, expanded data sampling, and potential pay equity audit measures to ensure fairness and mitigate strategic and compliance risks.

Employee Prediction Using SPSS Modeler

(Part B)

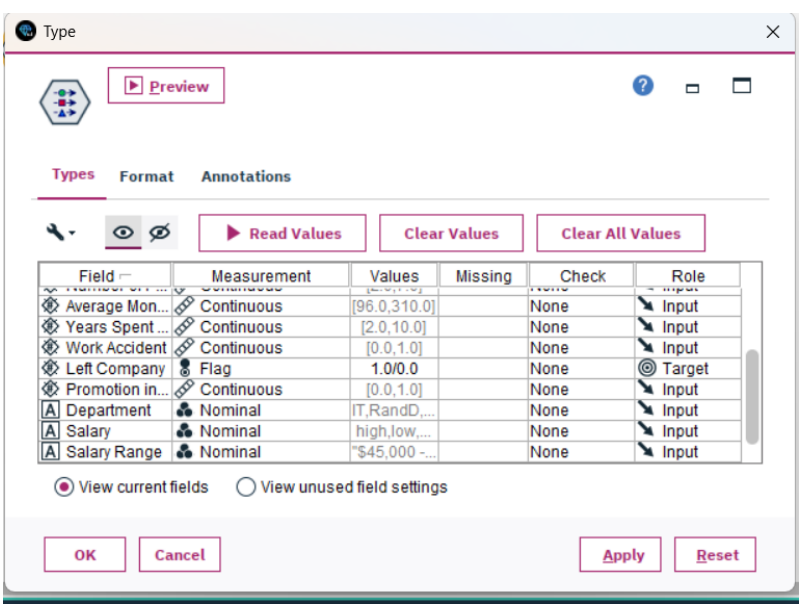
Randen Brown

1. INTRODUCTION

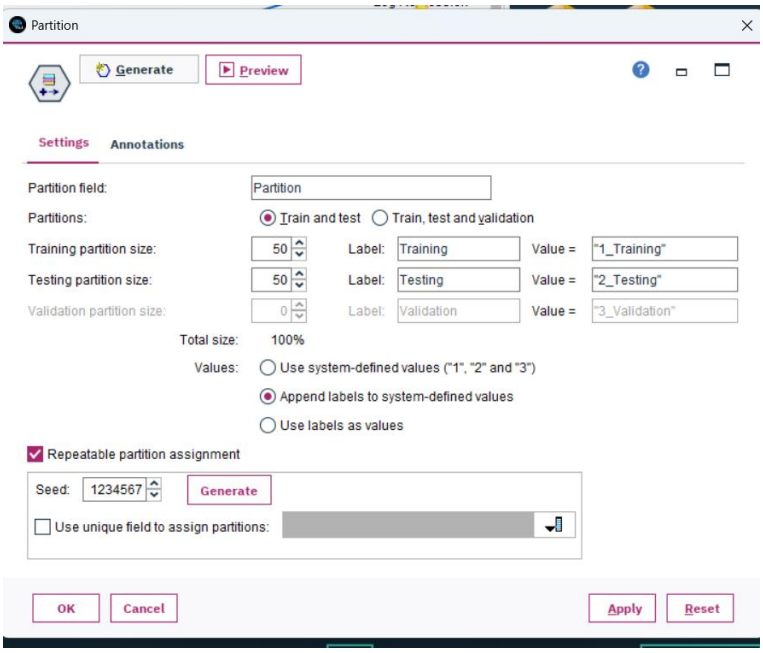
This project analyzes employee attrition using multiple machine learning approaches within IBM SPSS Modeler. Models include Logistic Regression, Feature Selection, K-Nearest Neighbors (KNN), Naïve Bayes, and Principal Component Analysis (PCA). The goal is to identify key drivers of employee turnover and compare predictive performance using a 50/50 training and testing partition.

2. DATA PREPARATION

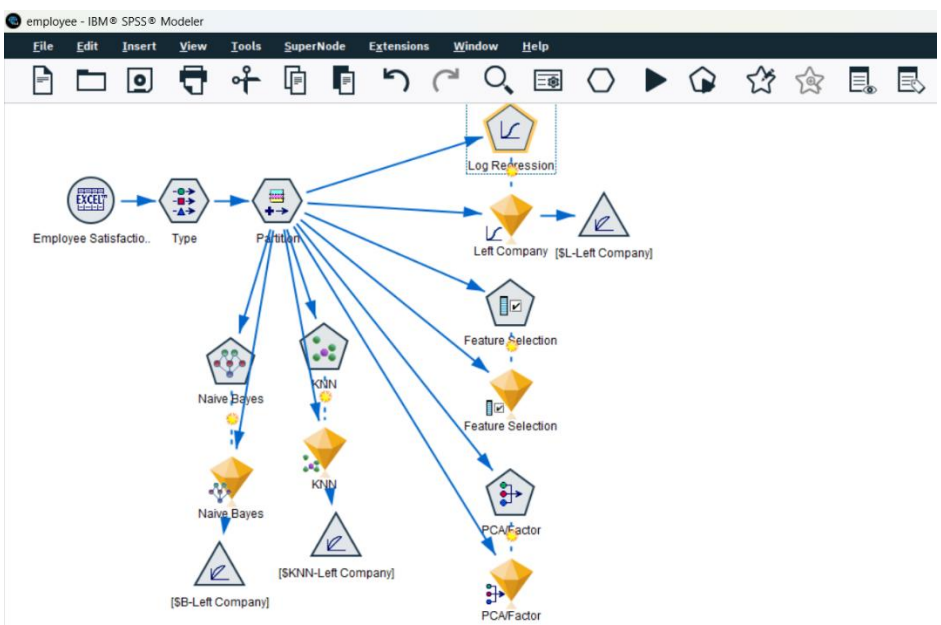
The dataset includes variables representing employee satisfaction, performance, workload, tenure, accidents, and promotions. Data was imported via the Excel Source node and variable roles were assigned using the Type node. The screenshot below displays the type node panel. It shows how each variable in the dataset is assigned a Measurement Level and Role. This step is important as SPSS requires every field to be correctly labeled before any modeling can occur. Left Company is set as the target variable since this is what is being predicted. Left company is also set as Flag because it will not be used in modeling.



Partition node used as told in the instructions is training: 50% and testing: 50%. This divides the data in half. Partitioning is a core step in predictive modeling because it enables the evaluation of model generalization.

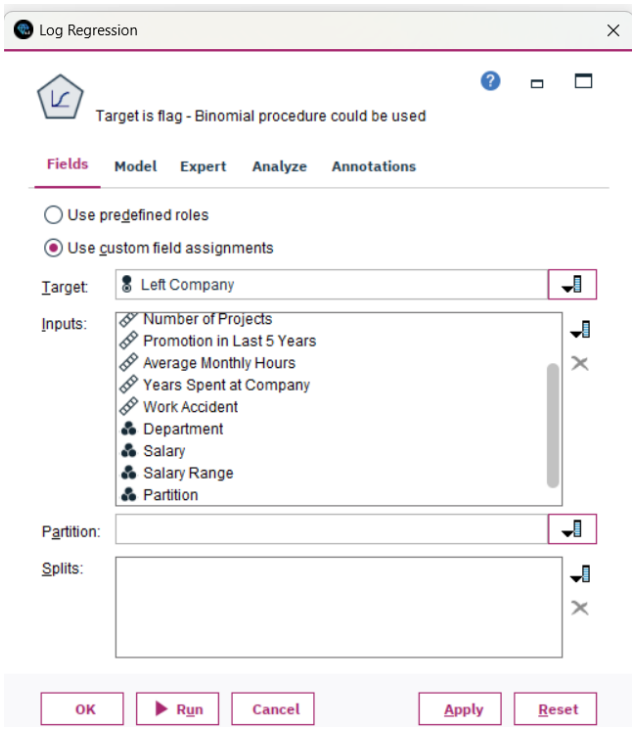


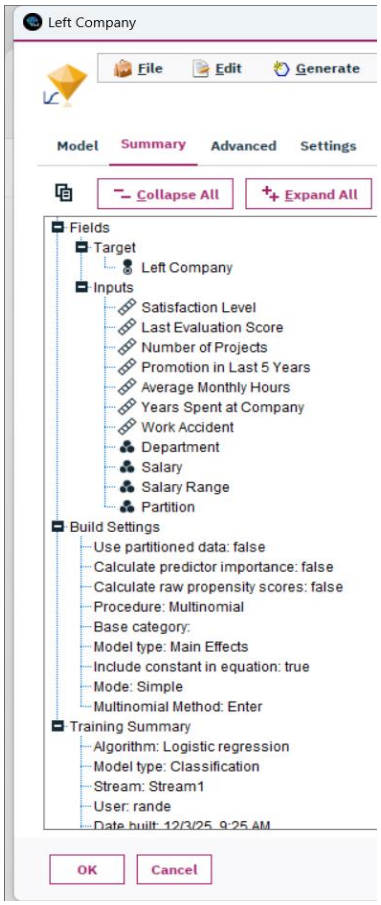
This is the final SPSS Modeler setup



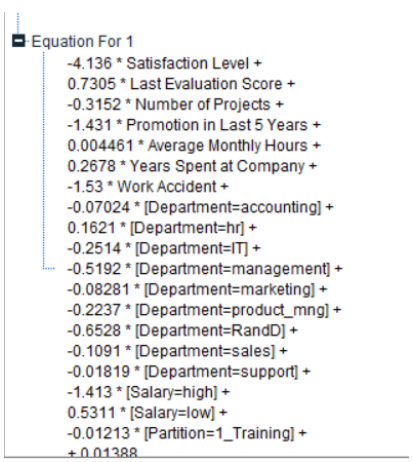
1. LOGISTIC REGRESSION MODEL

Model used all predictors with Left Company as the target. Logistic Regression served as the baseline model. The image below shows how the logistic regression model was configured. The target variable is Left Company, and all relevant attributes were included as inputs.





Below is a image of the logistic regression equation output. This provides the coefficients to each predictor to calculate the log-odds of "Left Compnay = 1". The numbers below represent coefficients showing how strongly it influences turnover. Negative coefficients reduce liklihood of leaving and positive coefficients increase likelihood of leaving. This is important because it provides direct interpretability of each variable impacts turnover.



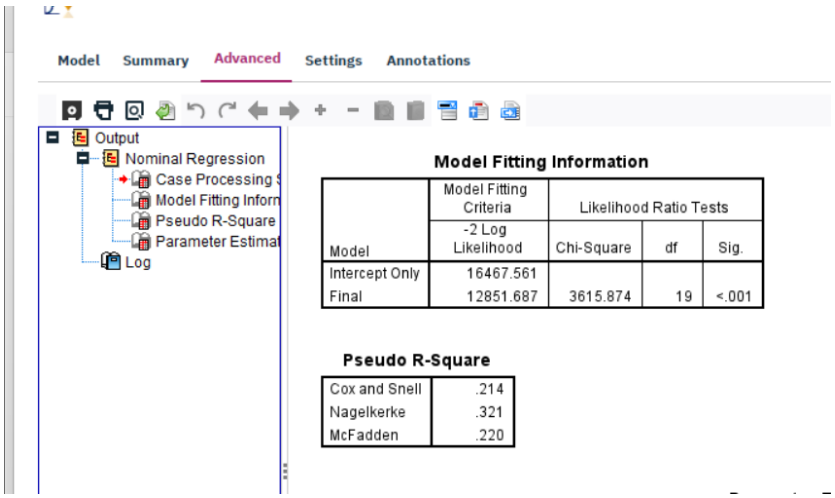
The company left counts had 11,428 (76.2%) stays and 3,572 (23.8%) left. This shows the dataset is imbalanced. Department frequencies show how many employees there are in HR, IT, Support etc. The salary range mirrors the salary distribution using income brackets. Finally, the partitions are 7,441 training and 7559 testing which is close to the 50/50 split. This table is used to validate that data has been properly prepared, there are no missing values exist.

Case Processing Summary

		N	Marginal Percentage
Left Company	0.0	11428	76.2%
	1.0	3572	23.8%
Department	accounting	767	5.1%
	hr	739	4.9%
	IT	1227	8.2%
	management	630	4.2%
	marketing	858	5.7%
	product_mng	902	6.0%
	RandD	787	5.2%
	sales	4140	27.6%
	support	2230	14.9%
	technical	2720	18.1%
Salary	high	1237	8.2%
	low	7317	48.8%
	medium	6446	43.0%
Salary Range	$45,000 - $74,999	6446	43.0%
	Greater than $75,000	1237	8.2%
	Less than $45,000	7317	48.8%
Partition	Training	7441	49.6%
	Testing	7559	50.4%
Valid		15000	100.0%
Missing		0	
Total		15000	
Subpopulation		13355[a]	

a. The dependent variable has only one value observed in 13355 (100.0%) subpopulations.

Below shows whether the model fits significantly better than the baseline. Some key findings are as follows. -2 Log Likelihood dropped from 16,467 to 12,851 and since the value is lower the fit is better. The Chi-square being 3615.874, $p < .001$ shows the model is statistically significant. The Pseudo and R-squared values indicate that the model explains about 21-32% variance, which is reasonable for HR turnover.



Model Fitting Information

Model	Model Fitting Criteria	Likelihood Ratio Tests		
	-2 Log Likelihood	Chi-Square	df	Sig.
Intercept Only	16467.561			
Final	12851.687	3615.874	19	<.001

Pseudo R-Square

Cox and Snell	.214
Nagelkerke	.321
McFadden	.220

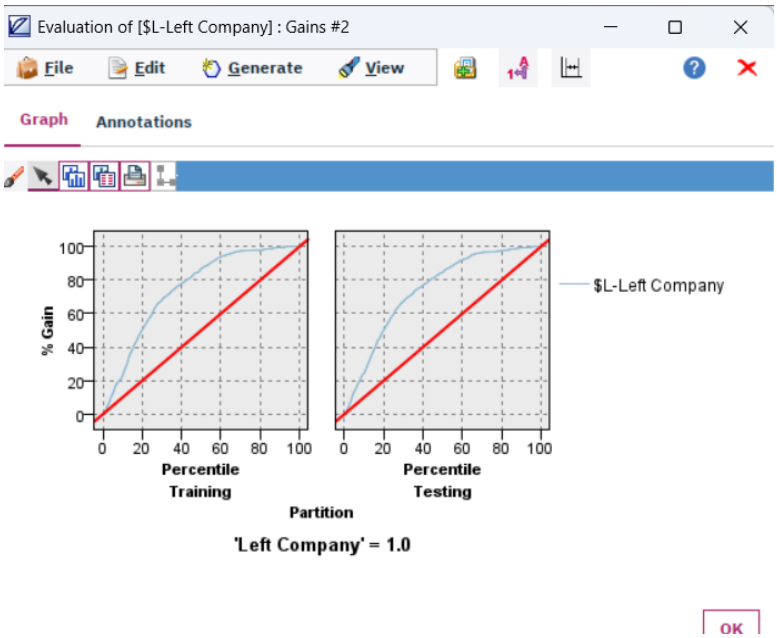
The parameter estimates table provides the coefficients, statistical significance, odds ratios, and confidence intervals for each predictor. Highly significant predictors are satisfaction level, last evaluation score, number of projects, promotion in last 5 years, years spent at company, work accident, several departments, and salary. The highest odds ratios belong to satisfaction level which proves it is a strong predictor and small increases reduce turnover.

Parameter Estimates

Left Company[a]		B	Std. Error	Wald	df	Sig.	Exp(B)	95% Confidence Interval for Exp (B)	
								Lower Bound	Upper Bound
1.0	Intercept	.014	.133	.011	1	.917			
	Satisfaction Level	-4.136	.098	1778.633	1	<.001	.016	.013	.019
	Last Evaluation Score	.730	.149	23.966	1	<.001	2.076	1.550	2.781
	Number of Projects	-.315	.021	218.510	1	<.001	.730	.700	.761
	Promotion in Last 5 Years	-1.431	.258	30.839	1	<.001	.239	.144	.396
	Average Monthly Hours	.004	.001	74.700	1	<.001	1.004	1.003	1.005
	Years Spent at Company	.268	.016	295.574	1	<.001	1.307	1.268	1.348
	Work Accident	-1.530	.090	291.952	1	<.001	.217	.182	.258
	[Department=accounting]	-.070	.107	.435	1	.510	.932	.756	1.149
	[Department=hr]	.162	.105	2.376	1	.123	1.176	.957	1.445
	[Department=IT]	-.251	.093	7.237	1	.007	.778	.648	.934
	[Department=management]	-.519	.140	13.833	1	<.001	.595	.453	.782
	[Department=marketing]	-.083	.106	.610	1	.435	.921	.748	1.133
	[Department=product_mng]	-.224	.104	4.655	1	.031	.800	.653	.980
	[Department=RandD]	-.653	.122	28.749	1	<.001	.521	.410	.661
	[Department=sales]	-.109	.066	2.765	1	.096	.897	.789	1.020
	[Department=support]	-.018	.076	.058	1	.810	.982	.847	1.139
	[Department=technical]	0[b]	.	.	0	.	.	.	.
	[Salary=high]	-1.413	.129	119.318	1	<.001	.243	.189	.314
	[Salary=low]	.531	.046	135.074	1	<.001	1.701	1.555	1.860
	[Salary=medium]	0[b]	.	.	0	.	.	.	.
	[Salary Range=$45,000 - $74,999]	0[b]	.	.	0	.	.	.	.
	[Salary Range=Greater than $75,000]	0[b]	.	.	0	.	.	.	.

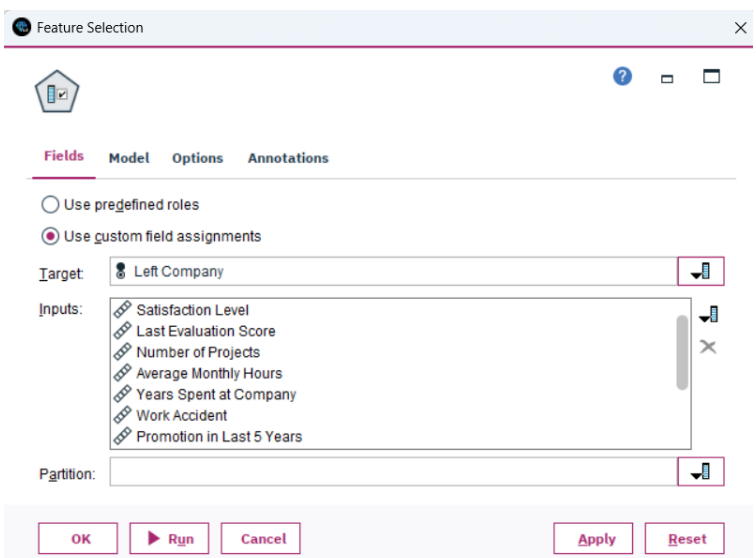
Evaluation:

The blue line represents the logistic regression model while the red line is a baseline random model. The logistic model performs far above random for both training and testing. The curve climbs sharply near the beginning meaning the model correctly identifies a large percentage of employees who leave in the top percentiles.



4. FEATURE SELECTION

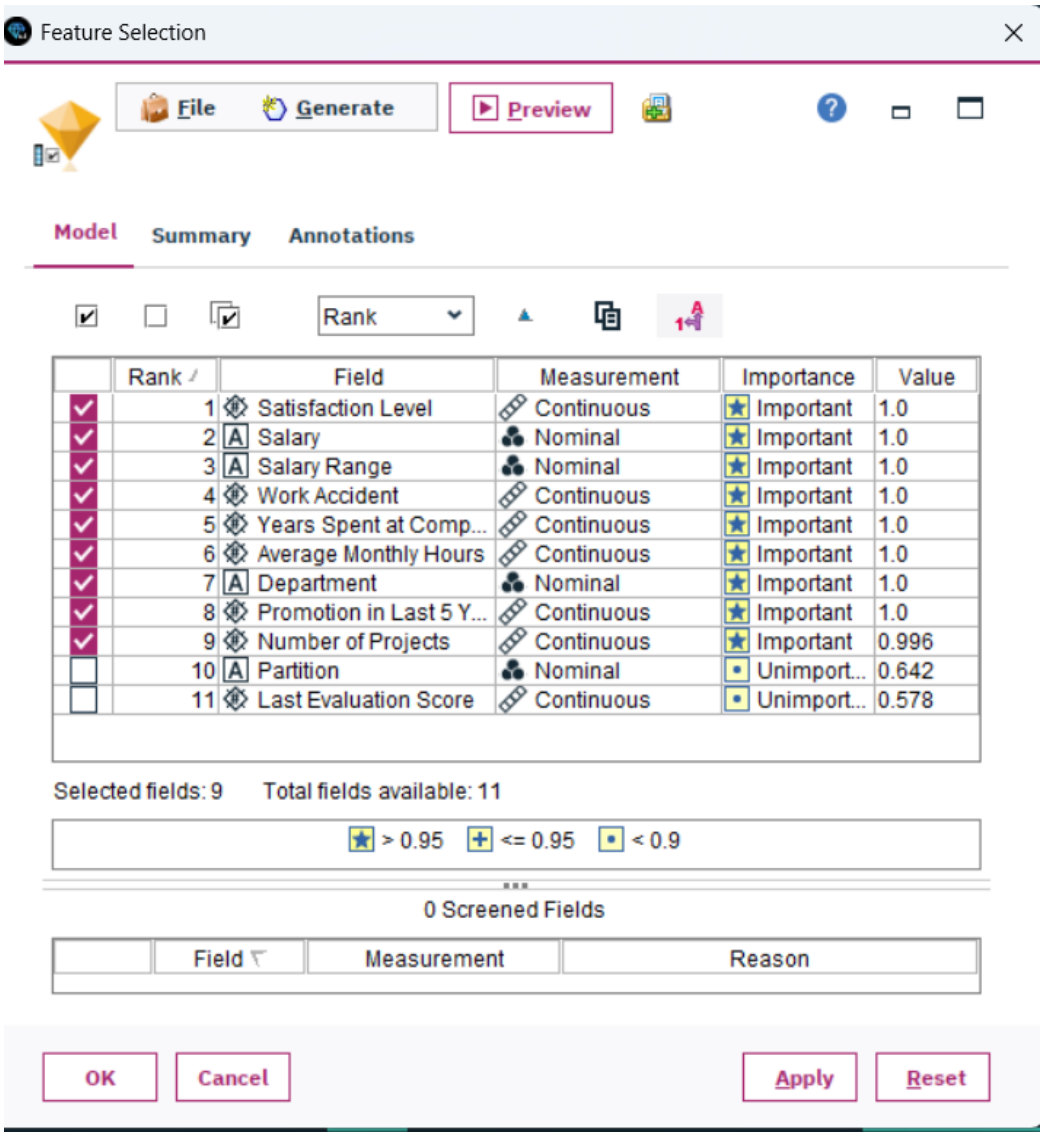
This shows the Fields tab of the feature selection node. The target is Left Company as we are trying to predict this value. Partition is empty because feature selection does not require it.



For the Model tab this was left at the default settings. This helps remove variables that have too many missing values or have no variation

This table shows where variables are ranked from most to least important. The scores range from 0 to 1 with 1 being the most important. The highest ranked fields are satisfaction level, salary, salary range, work accident, years spent at company, average

monthly hours, department, and promotion in last 5 years. The two unimportant variables are partition and last evaluation score. The table provides a clear hierarchy of predictor usefulness.

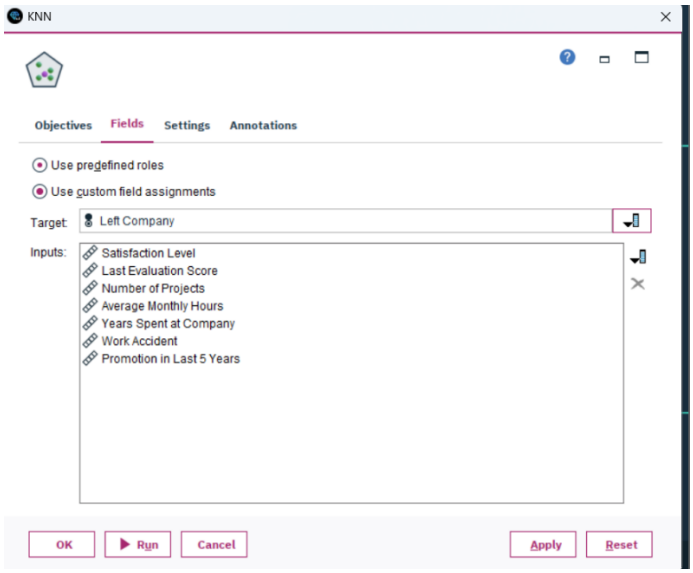


Rank	Field	Measurement	Importance	Value
1	Satisfaction Level	Continuous	Important	1.0
2	Salary	Nominal	Important	1.0
3	Salary Range	Nominal	Important	1.0
4	Work Accident	Continuous	Important	1.0
5	Years Spent at Comp...	Continuous	Important	1.0
6	Average Monthly Hours	Continuous	Important	1.0
7	Department	Nominal	Important	1.0
8	Promotion in Last 5 Y...	Continuous	Important	1.0
9	Number of Projects	Continuous	Important	0.996
10	Partition	Nominal	Unimport...	0.642
11	Last Evaluation Score	Continuous	Unimport...	0.578

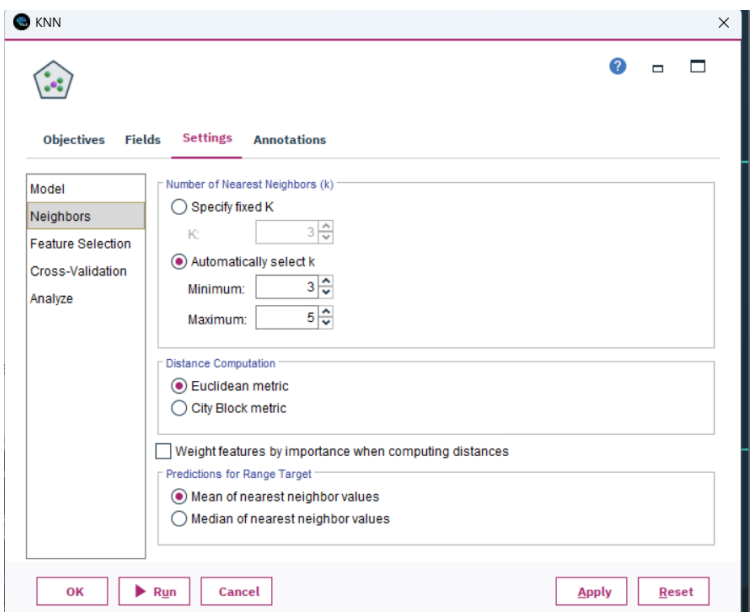
Field	Measurement	Reason

5. K-NEAREST NEIGHBORS (KNN) MODEL

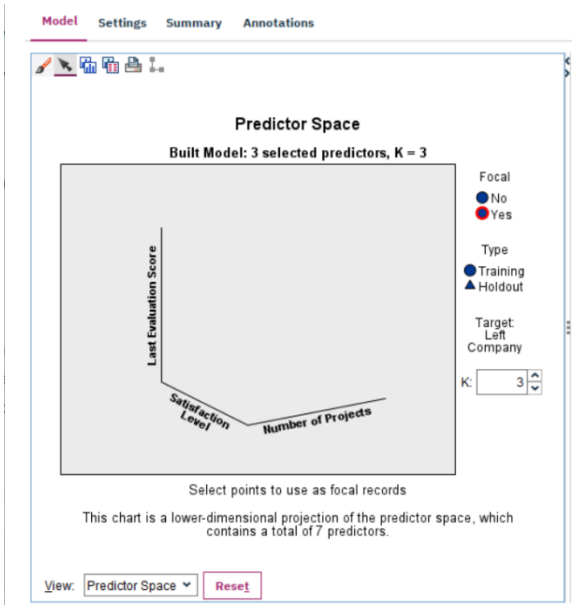
KNN used seven numeric predictors with Euclidean distance and automatic K selection. The target field is assigned to Left Company, meaning the model is predicting whether an employee leaves. The inputs are manually inserted, and no partition is needed at this step.



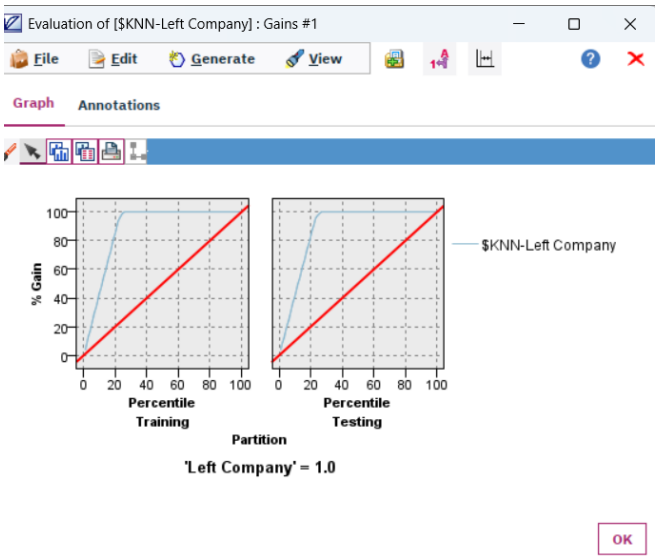
Below is the setting tab where the model is set to automatically select k with a minimum of 3 and maximum of 5. This allows the Modeler to test multiple values of k and choose the best-performing one. The Euclidean metric is selected as the distance measure, which is the standard approach for KNN models.



The predictor space is a plot that shows how selected predictors relate to each other in reduced space. The axes represent the predictors provided. The model summary indicates k = 3 was chosen as the number of neighbors.



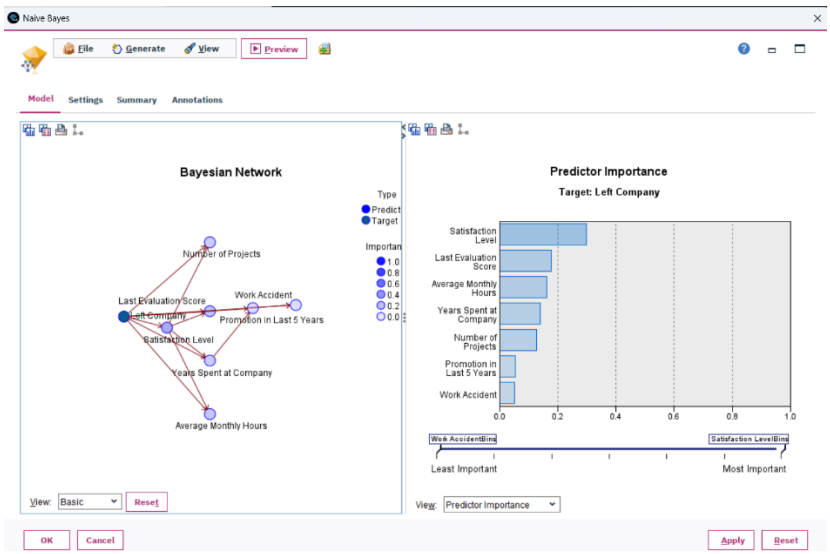
The blue curve represents model performance, while the red line represents random guessing. The model's gains curve rises sharply near the top, indicating strong predictive power. In the first 20% of the population scored, the model captures a large share of employees who left. The testing curve closely resembles the training curve, showing minimal overfitting.



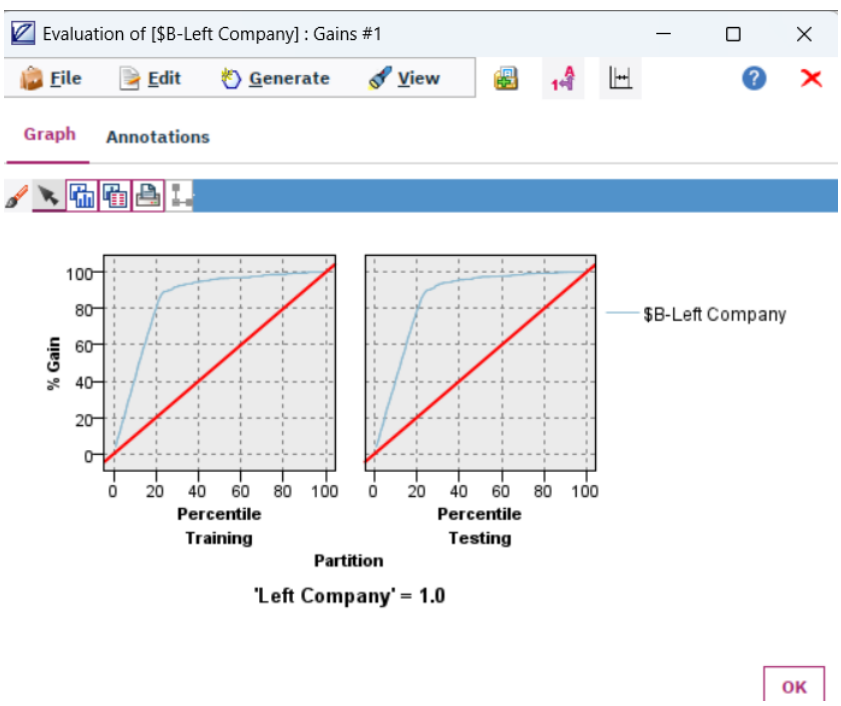
6. NAÏVE BAYES MODEL

This graph shows directional arrows indicating conditional dependencies. The Left Company is highlighted as the targeted. Predictors such a satisfaction level, last evaluation score, number of projects, and others are connected to the target and each other based on learned relationships. The representation shows how the Naïve Bayes

algorithm links variables. On the right is the predictor importance chart that ranks predictors by calculated importance. Satisfaction Level is shown as the most influential predictor. Other meaningful predictors include last evaluation score, average monthly hours, and years spent at company. This model shows the internal structure and how strongly each variable contribute to predicting employee turnover.

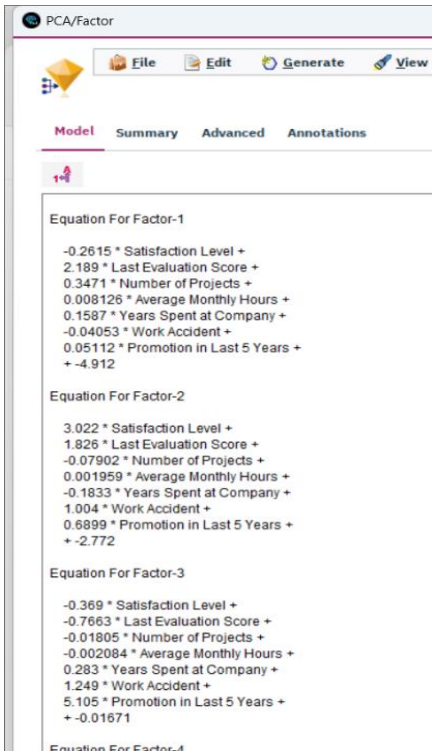


On the left graph the blue curve rises steeply, indicating that the model is effectively identifying employees who left. A step lift over the diagonal red line means strong performance. On the right graph the blue line is similarly steep, showing good generalization to unseen data. This confirms the Naïve Bayes model did not overfit.

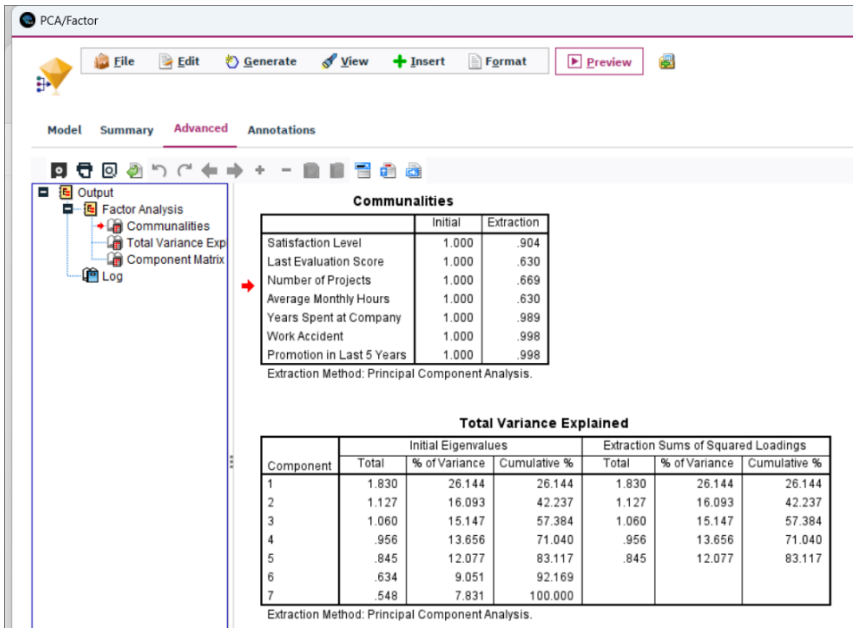


7. PRINCIPAL COMPONENT ANALYSIS (PCA)

This displays the component scoring coefficients generated by PCA. Each factor is weighted linear combination of the original input variables. Factor 1 has high positive weights for last evaluation score, number of projects, and years spent at company. Factor 2 is heavily influenced by satisfaction level and work accident, suggesting an employee-wellbeing dimension. Factor 3 loads strongly on promotion in last 5 years and years spent at company indicating promotion component,



The Extraction column shows how much of each variable's variance is captured by the extracted PCA components. A communality close to 1 indicates that PCA represents the variable very well. Component 1 shows 26.144% of total variance. Component 2 adds 16.093% totaling 42.237% cumulative. Component 3 adds 15.147% totaling 57.384% cumulative. Five components are enough to capture over 80% of the variance. This aligns with the standard PCA thresholds.



Communalities

	Initial	Extraction
Satisfaction Level	1.000	.904
Last Evaluation Score	1.000	.630
Number of Projects	1.000	.669
Average Monthly Hours	1.000	.630
Years Spent at Company	1.000	.989
Work Accident	1.000	.998
Promotion in Last 5 Years	1.000	.998

Extraction Method: Principal Component Analysis.

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	1.830	26.144	26.144	1.830	26.144	26.144
2	1.127	16.093	42.237	1.127	16.093	42.237
3	1.060	15.147	57.384	1.060	15.147	57.384
4	.956	13.656	71.040	.956	13.656	71.040
5	.845	12.077	83.117	.845	12.077	83.117
6	.634	9.051	92.169			
7	.548	7.831	100.000			

Extraction Method: Principal Component Analysis.

Component 1 shows a productivity dimension with number of projects and average monthly hours. Component 2 shows a satisfaction factor at .846. Components 3 and 4 appear to be connected by work accident. Component 5 has years spent at company as the strongest variable.

Component Matrix

	Component				
	1	2	3	4	5
Satisfaction Level	-.119	.846	-.097	.262	.309
Last Evaluation Score	.686	.352	-.139	.108	.068
Number of Projects	.783	-.110	-.024	-.101	-.181
Average Monthly Hours	.743	.110	-.110	-.010	-.231
Years Spent at Company	.424	-.301	.438	.039	.724
Work Accident	-.026	.398	.466	-.787	-.053
Promotion in Last 5 Years	.013	.112	.781	.495	-.362

Extraction Method: Principal Component Analysis.

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End of job:  42 command lines  0 errors  0 warnings  1 CPU seconds
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8. CONCLUSION

This analysis explored multiple predictive modeling techniques within IBM SPSS Modeler to examine the factors influencing employee turnover. Through feature selection, the most important drivers were identified as satisfaction level, salary, salary range, work accident, years spent at the company, and average monthly hours. PCA further supported these findings by revealing underlying patterns in the data, with the first

few principal components capturing the majority of variance. These components highlighted strong relationships between productivity metrics, employee satisfaction, and tenure, reinforcing the multidimensional nature of employee attrition.

Several machine learning models were developed and evaluated, including Logistic Regression, KNN, and Naive Bayes. Logistic Regression produced coefficients and strong statistical significance for key predictors such as satisfaction level, last evaluation score, number of projects, and work accident. KNN demonstrated strong predictive capabilities, particularly at lower percentiles, while Naive Bayes provided clear visual insights into predictor dependencies and importance. The gains charts for all models consistently showed performance well above the baseline diagonal line. Confirming that each model successfully captured meaningful patterns in the data and possessed strong predictive power. This project demonstrates that employee turnover is driven by a combination of performance indicators, satisfaction measures, workload, and organizational factors. By identifying the strongest predictors and validating them across several machine learning methods, this analysis provides a solid foundation for future predictive modeling